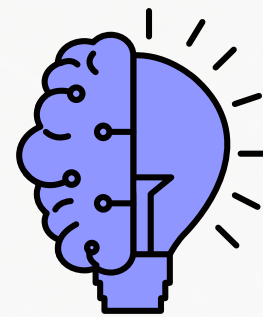


Agentic Customer Support System

MULTI-AGENT GRAPHRAG SYSTEM FOR INTELLIGENT
SUPPORT AUTOMATION



40-60%
Auto-Resolution



<15 sec
Response Time



24/7
Availability

Team 6: Data Alchemists

VASU BHUT
NISARG SHETH
GEETIKA BARLA

GITHUB REPO:

[HTTPS://GITHUB.COM/VAISU-
BHUT/DATAALCHEMIST](https://github.com/vaisu-bhut/dataalchemist)

The problem we are solving

Support Team Challenges

- ✗ Repetitive questions: "How do I reset password?" asked 50+ times/day
- ✗ Wasted time: Agents spend 5 minutes finding answers they've given before
- ✗ Knowledge loss: Solutions buried in old tickets, hard to find
- ✗ Limited coverage: No support during off-hours

Real Cost

50 repetitive tickets/day × 5 minutes each
= 250 minutes wasted daily
= 20+ hours per week

Our Solution: Three Key Ideas

1. Learn from History

Store past conversations in a **knowledge graph** that captures relationships between customers, issues, and solutions

2. Find Similar Issues

Use **vector embeddings** + **graph relationships** to find not just similar text, but similar problems with proven solutions

3. Generate Smart Answers

LLM synthesizes natural responses from multiple sources, only escalating when uncertain

System Architecture: Multi-Agent Design

FOUR SPECIALIZED AGENTS WORKING TOGETHER

Why Multi-Agent?

Each component scales independently. If chat volume spikes, we spin up more Chat Agents without touching anything else.

Master Agent

Routes requests,
coordinates workflow,
tracks progress

Ingest Agent

Processes
conversations, removes
PII, builds knowledge
graph

Chat Agent

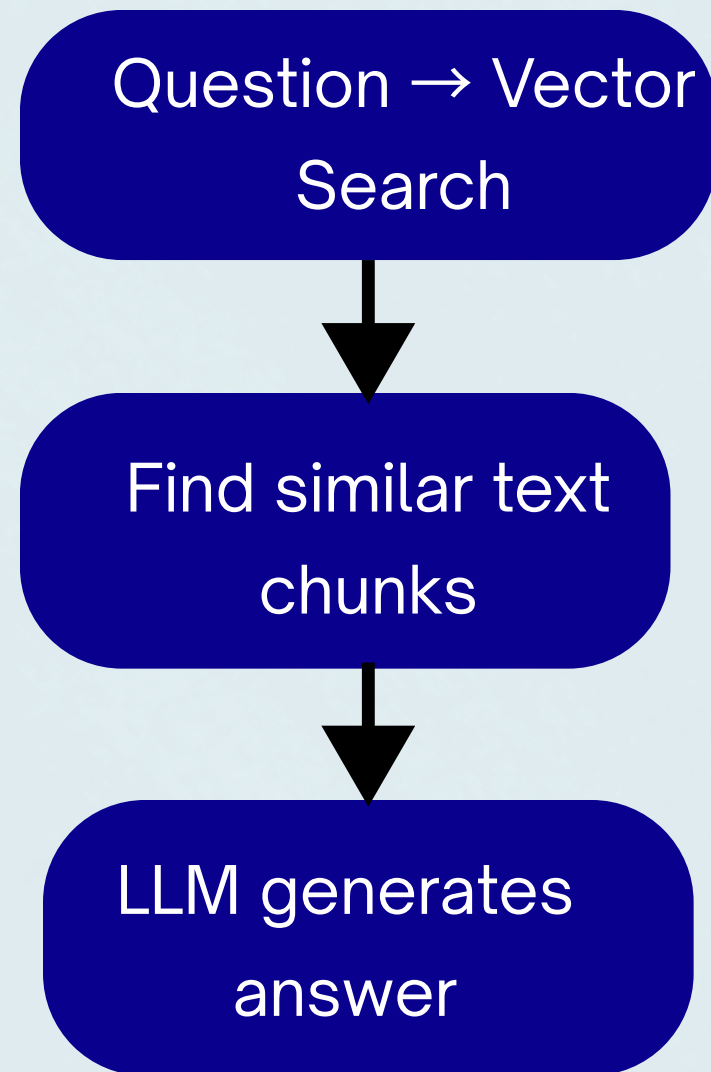
Answers questions
using vector search +
graph traversal

Analytics Agent

Tracks metrics, ROI,
system performance

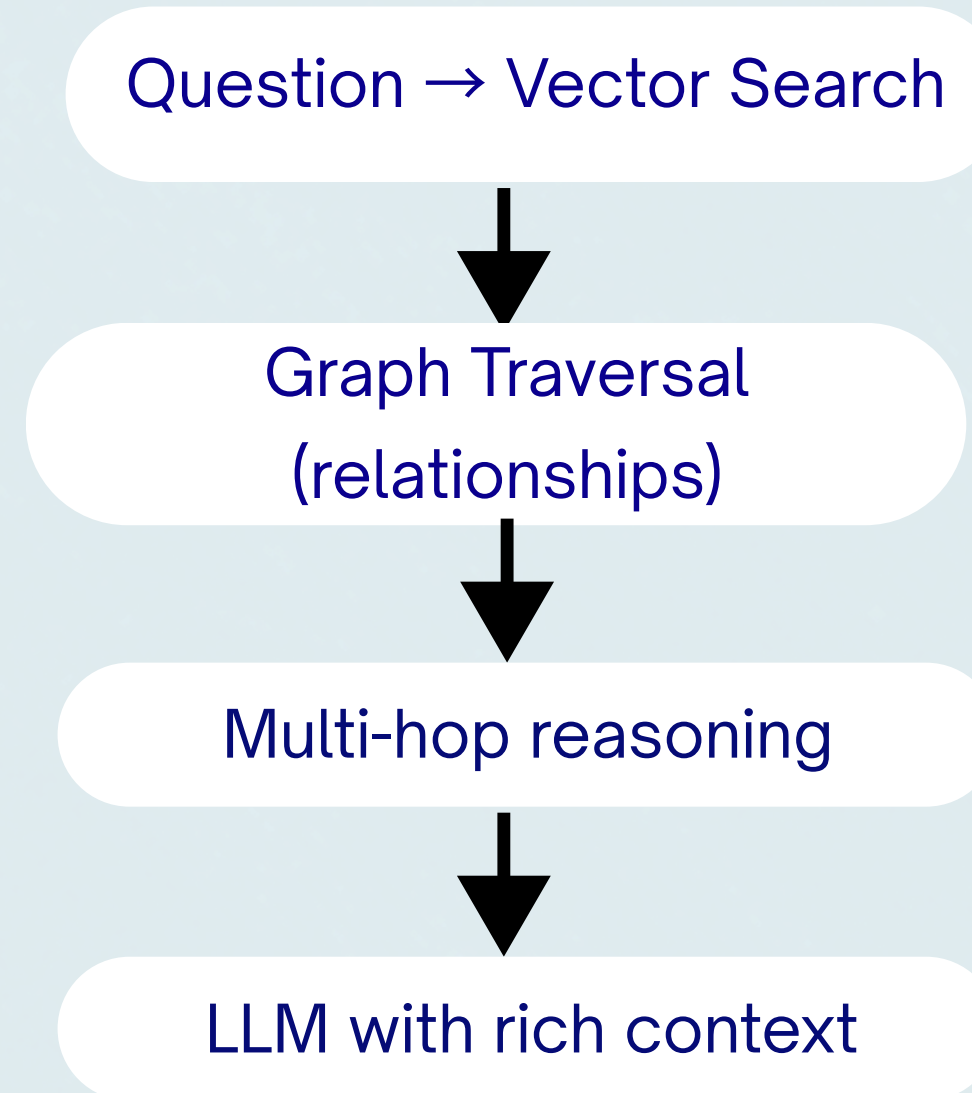
GraphRAG: Our Implementation

Traditional RAG



✗ Limited: Text similarity only

GraphRAG (Our Implementation)



✓ Smart: Context + relationships

Example: Instead of just "here's a similar issue", GraphRAG returns "here's a solution that worked for 12 similar issues across 3 customer types with 4.8/5 rating"

Impact

Measured Impact

40-60%
Questions
answered
automatically

20+ hrs
Weekly time
saved per team

5-15s
Average
response time

Thank You